CHAPTER 10

Vectors, straights and planes of space

S3, S6, S12, S15; synthesis S17

Before you start · Vectors and space reading

Without a calculator, in six minutes. Respond briefly. Mobilize Chasles, coordinates and collinearity
Then choose the targeted pick-up indicated under the questions.

1. Complete $\overrightarrow{AB} + \overrightarrow{BC} = \ldots$.

2. Calculate $\overrightarrow{AB}$ if $A(1;2)$ and $B(4;-1)$.

3. Vectors $(2;3)$ and $(-4;-6)$ Are they colinear?

4. Provide the midpoint of the $[AB]$.

5. Resolve 2 $x - 3 = 7$.

6. In a cube ABCDEFGH , E,F,G,H above A,B,C,D, name an edge parallel to (AB).

Responses and Remediation

$\overrightarrow{AC}$; $(3;-3)$; yes; $(2{,}5;0{,}5)$; $x = 5$. For example [DC], [EF] or [HG].

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Follow route A to B then B towards C; R04. **2.** Subtract the starting coordinates from those of arrival; R05. **3.** Search for the same coefficient in both coordinates; R06. **4.** Make the averages of the two coordinates ; R05. **5.** Add 3 then divide by 2, and check by substitution. **6.** Build and appoint the face of the cube; avoid concluding on the only perspective. Questions R serve as support the teacher, even if their collective ritual is planned later.

Objectives

Calculate with vectors; use collinearity and coplanarity; characterize straight and planes; work in a coordinate system.

Course notes

Vector calculation

Chasles' relationship $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$, the sum of vectors and the product by a real extend
Two non-zero vectors are colinear if one is a multiple of the other.
Three vectors are coplanar if they belong to the same plane
Yes, I am. $\vec{u}, \vec{v}$ are not colinear, $\vec{u}, \vec{v}, \vec{w}$ are coplanar if and only if $\vec{w} = \alpha\vec{u} + \beta\vec{v}$ for some real α, β.

Right and plan

The right through A of a non-zero master vector $\vec{u}$ is all points M such that $\overrightarrow{AM} = t\vec{u}$ for a real t. The plan passing through A and directed by two non-colonial vectors $\vec{u}, \vec{v}$ is all points M such that $\overrightarrow{AM} = s\vec{u} + t\vec{v}$ for real s, t.

Three non-aligned items define a plan. $A, B, C, D,$ A, B, C Non-Aligned coplanar if and only if $\overrightarrow{AD}$ is linear combination of $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

Positions

Two straight lines are confused, strictly parallel, seizing or not coplanar.
colinear directors characterize parallel straights in the broad sense.
do not provide the intersection. A straight and a plane: straight included in the plane, parallel and disjointed,
Two planes: combined, strictly parallel, or slatting according to one point.
Right.

Example in a cube. In ABCDEFGH , the faces $ABCD$ and $EFGH$ are parallel and edges AE, BF, CG, DH The lines match. (AB) and (CG) are neither parallel nor secant: they are non-coplanar.

Base and coordinates

A base of space is a family of three non-coplanar vectors.
unique way $x\vec{\imath} + y\vec{\jmath} + z\vec{k}$. A reference is made up of an origin O and a base.

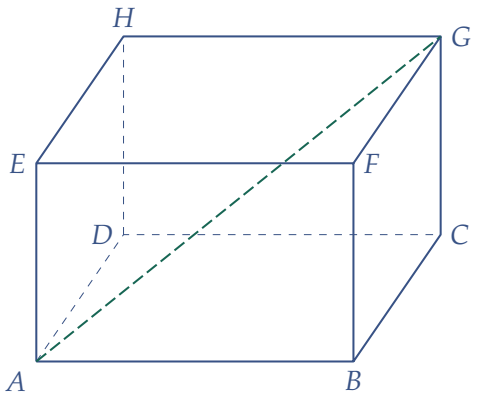
$$\overrightarrow{AB} = (x_B - x_A, y_B - y_A, z_B - z_A), \qquad I = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}, \frac{z_A + z_B}{2}\right).$$

In an orthonormal coordinate system, $\|\vec{u}\| = \sqrt{x^2 + y^2 + z^2}$ and $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$.

Example. In an orthonormal coordinate system, $A(1, 0, 2)$, $B(3, -1, 4)$, $\overrightarrow{AB} = (2, -1, 2)$, $AB = 3$ and midpoint is $(2, -\frac{1}{2}, 3)$.

Test a linear combination

Write the three co-ordinates. Determine coefficients from two equations
When possible, then check the third. A contradiction proves non-membership;
We don't remove the awkward equation.



$[AG]$ is a diagonal of space, plotted in green dots.
The hidden edges are dotted blue; D and F are not sure (AG).

Build a reliable representation

10.1 Combinations and any base

First use: S3 and S6.

Definition · Decompose a vector

A base of space is a triplet of non-coplanar vectors.
a unique way as a combination of these three vectors. A base is not necessarily
orthogonal or formed by unit vectors.

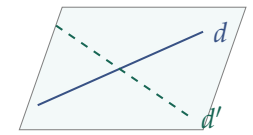
Example solved · In a tetrahedron

In a tetrahedron not flattened $OABC$, posisons $\vec{u} = \overrightarrow{OA}$, $\vec{v} = \overrightarrow{OB}$, $\vec{w} = \overrightarrow{OC}$. If I is the midpoint of $[AB]$, then $\overrightarrow{OI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v}$. So $\overrightarrow{CI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v} - \vec{w}$. These equalitys use Chasles and the midpoint, not lengths or angles.

A1 · Immediate application With the same base, J is the midpoint of $[OC]$ and K checks $\overrightarrow{OK} = 2\vec{u} - \vec{v}$. Express $\overrightarrow{JK}$ then $\overrightarrow{IK}$.

10.2 See not enough to conclude

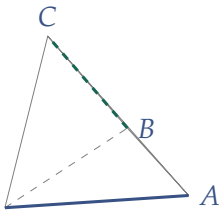
First use: S12.



pruning, same plane



parallel, same plane



non-coplanar edges

Property · Two-right positions

Two lines are either coplanar (secting or parallel) or non-coplanar.
non-collinear managers exclude parallelism, but do not prove an intersection.

Example Solved · Two non-coplanar lines

In tetrahedron $OABC$, the right (OA) and (BC) are not coplanar. If a plan
both contained, it would contain O, A, B, C, which would contradict the tetrahedron hypothesis
not flattened. An intersection of their lines on the sheet would only be a projection effect.

A2 · Read and argue On the three diagrams, name the position of the straights.
geometrically accompanying the drawing to allow this answer ? Give an example of straight lines not
coplanar in a cube ABCDEFGH .

A3 · Refuter "Two straight lines whose guiding vectors are not colinear are pruning."
Refuse with tetrahedron, then add a hypothesis that makes the statement true.

10.3. Characterize a plane by two directions

First use: S12 and S15.

Example resolved · Membership and uniqueness

On a base, be $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\overrightarrow{AM} = (2;3;5)$. We're looking. λ, μ such that $\overrightarrow{AM} = \lambda\vec{u} + \mu\vec{v}$. The first two coordinates require $\lambda = 2$, $\mu = 3$ and the third check $5 = 2 + 3$. Item M belongs to the point plan A and Directions $\vec{u}, \vec{v}$. For $(2;3;4)$, the The last tie fails: the point does not belong to it.

A4 · Partial guidance The vectors $\vec{u} = (1;1;0)$ and $\vec{v} = (0;1;1)$ Are they colinear? value of m the vector $(2;m;3)$ Is it a combination of $\vec{u}$ and $\vec{v}$?

A5 · Correct an error On any basis, a student calculates the norm of the vector of Coordinated $(1;1;0)$ by $\overline{2}$. Explain the missing hypothesis. Give an example with two unit vectors at an angle of 60 °.

10.4. Moving from solid to calculation

First use: S17.

Example Solved · Middles and Parallelism

If I and J are the midpoints of [OA] and [OB], $\overrightarrow{IJ} = \overrightarrow{IO} + \overrightarrow{OJ} = -\frac{1}{2}\overrightarrow{OA} + \frac{1}{2}\overrightarrow{OB} = \frac{1}{2}\overrightarrow{AB}$. The rights (IJ) and (AB) therefore have colinear directions. This calculation is valid in any triangle not flattened.

A6 · Choice of strategy In tetrahedron $OABC$, I is the midpoint of $[OA]$, J that of $[OB]$ and K the of $[OC]$. Demonstrate that plans (IJK) and (ABC) are parallel, then that O does not belong to (IJK). Do not use Cartesian equation.

A first step: Compare $\overline{IJ}$ to $\overrightarrow{AB}$ and $\overrightarrow{IK}$ to $\overrightarrow{AC}$. For membership, decompose in $(\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC})$.

Wait a minute at the courthouse. Describe the tetrahedron and the three midpoints without showing the figure. must be able to place them. Specify the segments rather than the "left-hand point".

Review postponed · S51 and then S70 The same plane will be described by a normal vector, then an equation. Vector definition remains the means of controlling the calculation.

Research activity

Three midpoints in a cube · 40 min.

The cube is considered ABCDEFGH shown above. $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are the respective circles of $[AB]$, $[AD]$ and $[AE]$. We note O the center of the cube.

1. Express $\overline{IJ}$ and $\overrightarrow{IK}$ in base $(\vec{\imath}, \vec{\jmath}, \vec{k})$.

2. Demonstrate points I, J and K are not aligned.

3. Determine whether O belongs to the plan (IJK). The answer will be justified by whether $\overrightarrow{IO}$ is a linear combination of $\overline{IJ}$ and $\overrightarrow{IK}$.

4. Why does the single reading of the figure not allow us to demonstrate belonging or non-membership O on the plan (IJK) ?

Progressive exercises

E1 · Direct application. In a base of space, vectors $\vec{u}$ and $\vec{v}$ have for coordinates respective $(1;2;-1)$ and $(3;6;-3)$. Are they colinear? Justify.

E2 · Application and justification. In an orthonormal coordinate system of space, we consider $A(0;1;2)$ and $B(4;-1;0)$. Determine the coordinates of the midpoint of the $[AB]$then calculate the distance AB.

E3 · Method to be organised. In a base of space, we consider $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\vec{w} = (2;3;5)$. Express $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$, and then deduce that these three vectors are coplanar.

E4 · Choice of strategy. In a base of space, vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ have for coordinates $(1;0;1)$, $(0;1;1)$ and $(1;1;0)$. The triplet $(\vec{u},\vec{v},\vec{w})$ Is it a base of space? Justify.

E5 · Synthesis or demonstration. In a space coordinate system, we consider $A(1;0;2)$, $B(3;-1;4)$ and $C(5;-2;6)$. Determine whether C belongs to the right (AB). Justify by vector calculation.

Work in the manual or annal

Barbazo, p. 328, exercise 21. In class after S15: address the three questions. For each intersection, begin by identifying each of the plans from the tops of the cube and the indicated environments. Distinguish two distinct planes, confused and strictly parallel.

Duty at home

DM10 · Points of a tetrahedron · 50 min.

A tetrahedron is considered $ABCD$ not flattened. Points I, J and K are the respective circles of $[AB]$, $[AC]$ and $[AD]$.

1. Express $\overline{IJ}$ according to the $\overrightarrow{BC}$ and $\overrightarrow{IK}$ according to the $\overrightarrow{BD}$.

2. Deducing that plans (IJK) and (BCD) are parallel.

3. Define point G by $\overrightarrow{AG} = \frac{1}{4}(\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD})$. The point G Does it belong to the plan? (IJK) ? Justify.

4. For this question alone, it is assumed that, in an orthonormal coordinate system, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$ and $D(0;0;4)$. Calculate lengths IJ, IK and JK.

MCQ A · consolidation (after S6)

After S6.

For each question, only one proposal is accurate. consult the correction.

1. On a base, $(2;4;-2)$ and $(1;2;-1)$ are:

A. orthogonal

B. Colinear

C. Non-coplanar

D. Equal

2. Three vectors $\vec{u}, \vec{v}, \vec{w},$ $\vec{u}, \vec{v}$ co-planar if:

A. $\vec{w} = \alpha\vec{u} + \beta\vec{v}$

B. $\vec{w} = 0$ only

C. Their norms are equal

D. they have three coordinates

3. Two non-collinear master vector space lines are necessarily:

A. secantes

B. Non-coplanar

C. either seizing or not coplanar

D. Parallels

4. In a database, writing a vector is:

A. Always impossible

B. unique when the three vectors form a single base

C. Independent of the base

D. For orthonormal markers only

5. A ridge hidden from a perspective must be:

A. deleted

B. confused with a diagonal

C. indicated by a different trait

D. declared non-existent

QCM B · Geometric review (after S17)

After S17.

For each question, only one proposal is accurate.
consult the correction.

6. To show that $M \in (ABC)$where: A, B, C are not aligned, we can:

A. Compare AM to AB

B. Writing $\overrightarrow{AM} = s\overrightarrow{AB} + t\overrightarrow{AC}$

C. calculating an area only

D. Look at the figure

7. If two planes have the same direction, they are:

A. perpendicular

B. Parallel or combined

C. necessarily confused

D. secants

8. In an orthonormal coordinate system, the midpoint of $A(0;1;2)$ and $B(2;3;4)$ is:

A. $(1;2;3)$

B. $(2;4;6)$

C. $(1;1;2)$

D. $(0;2;3)$

9. Two lines whose projections cut on the sheet:

A. are pruning

B. are parallel

C. may be non-coplanar

D. are confused

10. For a triplet to be a base of space, just check:

A. Non-coplanarity

B. Orthogonality

C. Equality of norms

D. Origin of vectors

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Two non-parallel space lines shall be skewed.

2. The null vector is colinear to any vector.

3. A figure in perspective can serve as evidence of co-planarity.

Review review

I can choose a base, calculate a linear combination and distinguish relative positions. **To be monitored:**
take an appearance of the figure for a space property; forget an equation of coordinates.

Corrected MCQs

1. **B.** The first vector is twice the second. Their scalar product is not available in any base; collinearity does not depend on a metric.
2. **A.** With the first two non-collinear vectors, the linear combination characterizes their plane The null vector is a particular case, not the only one.
3. **C.** Non-collinear directors exclude parallelism and confusion. The intersection remains to be established: the lines may also be non-coplanar.
4. **B.** A basis guarantees the existence and uniqueness of decomposition. the basics, which do not need to be orthonormized.
5. **C.** The interrupted trait retains the edge and indicates that it is hidden. Clearing the edge would remove information on the solid.
6. **B.** The directions AB, AC the direction of the plan. A relationship $AM = sAB + tAC$ therefore proves belonging.
7. **B.** Two planes of the same direction are parallel in the broad sense. distinguish planes combined and strictly parallel.
8. **A.** We average each pair of coordinates: $(1;2;3)$. This midpoint formula does not does not even require orthonormality.
9. **C.** The perspective is a projection that loses depth. An intersection drawn does not prove not an intersection of the space lines.
10. **A.** Three non-coplanar vectors form a basis. Orthogonality and norm 1 are not required.

True/false. False: the straight (AB) and (CG) a cube is not coplanar. True: $\vec{0} = 0\vec{u}$. False: a projection can create apparent intersections; a calculation or geometric proof is necessary.

Corrected applications and support

Application 1. $\overrightarrow{JK} = 2\vec{u} - \vec{v} - \frac{1}{2}\vec{w}$; $\overrightarrow{IK} = \frac{3}{2}\vec{u} - \frac{3}{2}\vec{v}$.

Application 2. The first two drawings place the right in a common plane; the third one represents opposite edges of an unflatted tetrahedron. In a cube, (AB) and (CG) are no coplanar: the only plan containing A, B, C is that of the face $(ABCD)$which does not contain G.

Implementation 3. (OA) and (BC) Add "coplanar". Two straight lines distinct from the same plane, not parallel, are stinging.

Implementation 4. The first coordination requires that a possible relationship between $\vec{v}$ and $\vec{u}$ either 0, impossible since $\vec{v} \neq 0$. The combination is $(\lambda; \lambda + \mu; \mu) : \lambda = 2$, $\mu = 3$, so $m = 5$.

Application 5. It takes an orthonormal coordinate system. With two unit vectors of angle 60 °, the square of the norm of their sum is $1 + 1 + 2\cos 60° = 3$; the norm is $\overline{3}$. In the first passage, you can also see this length as a diagonal of diamond.

Implementation 6. $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AB}$ and $\overrightarrow{IK} = \frac{1}{2}\overrightarrow{AC}$: two non-collinear directions of the planes are parallel, So the planes are parallel. $O \in (IJK)$, then $\overrightarrow{IO} = \lambda\overrightarrow{IJ} + \mu\overrightarrow{IK}$. The coefficients $\overrightarrow{OB}$ and $\overrightarrow{OC}$ require $\lambda = \mu = 0$, while that of $\overrightarrow{OA}$ would be $-\frac{1}{2} = 0$, contradiction.

Space research

Manipulate four non-coplanar points, then draw. In the correction, distinguish this to provide assistance, to provide three Chasles decompositions; to deepen, replace the midpoints with the same ratio $t \in]0; 1[$ and find parallelism.

Conduct and correction

Benchmarks for the teacher

The DS1 precedes S12 and S15: propose vector geometry in a given base, without require new coordinate techniques. Use a material cube, then a perspective cavalier; mark the hidden edges. Two non-parallel lines of space are not The norm by coordinates requires an orthonormal coordinate system. formalized parametric representations and plan equations will be processed later.

Corrected activity

1. By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{\jmath}$ and $\overrightarrow{IK} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{k}$.

2. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would be $\frac{1}{2} = 0$, impossible in one base. These two vectors are therefore non-collinear; I, J, K define a plan.

3. The center of the cube checks $\overrightarrow{AO} = \frac{1}{2}(\vec{\imath} + \vec{\jmath} + \vec{k})$, therefore $\overrightarrow{IO} = \frac{1}{2}\vec{\jmath} + \frac{1}{2}\vec{k}$. If $O \in (IJK)$, there is s, t such that $\overrightarrow{IO} = s\overrightarrow{IJ} + t\overrightarrow{IK}$. By comparing the base coefficients,

$$0 = -\frac{s+t}{2}, \qquad \frac{1}{2} = \frac{s}{2}, \qquad \frac{1}{2} = \frac{t}{2}.$$

The last two require $s = t = 1$, while the first would ask $s + t = 0$. Contradiction : $O \notin (IJK)$.

4. The cavalier perspective is a projection on a sheet: separate points may have the It helps to conjecture, but an apparent superimposition does not prove belonging to a plan.

E1. The three coordinates of $\vec{v}$ are worth three times that of $\vec{u}$: $\vec{v} = 3\vec{u}$. The two vectors are therefore Collinear.

E2. The midpoint has its coordinates $(\frac{0+4}{2}, \frac{1-1}{2}, \frac{2+0}{2}) = (2, 0, 1)$. The difference in coordinates gives $\overrightarrow{AB} = (4, -2, -2)$; in the orthonormal coordinate system, $AB = \sqrt{16 + 4 + 4} = 2\sqrt{6}$.

E3. We calculate $2\vec{u} + 3\vec{v} = (2, 0, 2) + (0, 3, 3) = (2, 3, 5) = \vec{w}$. Thus $\vec{w}$ belongs to the vector plan led by the first two vectors: the three are coplanar.

E4. The first two vectors are non-colonial. If the third was their combination $\lambda(1, 0, 1) + \mu(0, 1, 1)$, the first two coordinates would require $\lambda = \mu = 1$, but the third would be worth $2 \neq 0$. The three vectors are non-coplanar and thus form a base of space.

E5. $\overrightarrow{AB} = (2, -1, 2)$ and $\overrightarrow{AC} = (4, -2, 4) = 2\overrightarrow{AB}$. Like $A \neq B$, right (AB) is defined and this collinearity proves $C \in (AB)$.

Corrected from DM10

1. By Chasles and the definitions of midpoints,

$$\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AC} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BC}, \qquad \overrightarrow{IK} = \frac{1}{2}\overrightarrow{AD} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BD}.$$

2. Points B, C, D are not aligned because the tetrahedron is not flattened. $\overrightarrow{BC}, \overrightarrow{BD}$ are therefore non-colonial, as well as $\overrightarrow{IJ}, \overrightarrow{IK}$. Plans have the same direction. They are distinct: if $I \in (BCD)$, right $(BI) = (BA)$ would be included in this plan, which would place A, contradiction. They are therefore strictly parallel.

3. In the base $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$, we have $\overrightarrow{IG} = (-\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $\overrightarrow{IJ} = (-\frac{1}{2}, \frac{1}{2}, 0)$ and $\overrightarrow{IK} = (-\frac{1}{2}, 0, \frac{1}{2})$. A relationship $\overrightarrow{IG} = s\overrightarrow{IJ} + t\overrightarrow{IK}$ imposes $s = t = \frac{1}{2}$ by the last two coordinates; the first would then be $-\frac{1}{2}$, and not $-\frac{1}{4}$. So $G \notin (IJK)$.

4. In this particular case, $I(2,0,0)$, $J(0,2,0)$ and $K(0,0,2)$. We find $IJ = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}$and of Same $IK = JK = 2$ $\overline{2}$. The triangle is equilateral. This metric conclusion does not apply to any non-flatted tetrahedron: it uses the coordinates of the particular case.

Corrected work in the manual or an anal

In this exercise, I is the midpoint of $[AB]$ and J that of $[CD]$.

1. Plans (AIE) and (BIG) both contain (AB). They are distinct: G does not belong to the Plan of the face $ABFE$. Their intersection is therefore exactly the right (AB).

2. Points A, D, I are three non-aligned points on the face $ABCD$; the same applies to B, J, C. The two plans are therefore confused with (ABC). The intersection is the integer plane, not a straight one.

3. The plan (HEF) is that of the upper face; (BJC) is that of the lower face. Strictly parallel: their intersection is empty.